

EE 584: Power System Protection

Course Code	EE 584
Course Name	Power System Protection
L-T-P-C	3-0-0-3
Prerequisites	EE-303 (Power systems)
Intended for	UG/PG
Distribution	Elective for B. Tech (EE), M.S., M. Tech. & Ph.D.
Approval	recommended by the 37th BoA (Oct 2020)

Course Contents

- **Introduction:** Fundamentals of protection- Security, selectivity and reliability, measurement principles unit and non-unit protection, legacy relays, solid state and numerical relays, standards in power system protection. [3 Lectures]
- **Instrument Transformers:** Working principle of current transformers (CT), voltage transformers (VT), dynamic response of CTs and capacitor coupled voltage transformer (CCVT) during faults and its effect on relaying. [4 Lectures]
- **Fault Analysis using Symmetrical components :** Sequence components, sequence modelling of power system components such as transformers, generators, transmission lines, fault analysis. [4 Lectures]
- **Numerical Relaying:** Sampling of analog values, analog to digital conversion, least square method for estimation of phasors, Fourier analysis, discrete Fourier transform: properties, phasor calculation, fast Fourier transform. [4 Lectures]
- **Directional Overcurrent Protection:** Directional and overcurrent relay principles, directional relay coordination problem associated with multiple loop system. [4 Lectures]
- **Transmission line Protection:** Distance and non-distance-based protection concepts. Zones of protection and back up protection, distance relay settings, pilot protection with distance relays. [6 Lectures]
- **Power Swing:** [2 hours] Power swing detection, stable and unstable swing, blocking and unblocking of distance relays during power swing, analysis of power swing in multi machine system, operation of relays during out of step condition. [2 Lectures]
- **Transformer protection:** Inrush phenomenon, Inrush detection methods; Differential and over-excitation protection. [3 Lectures]
- **Bus Protection:** Bus configurations; High and low impedance protection concepts. External Fault detection methods and remedial measures to account for CT saturation detection. [3 Lectures]
- **Rotating Machinery Protection:** Motor and generator protection, generator construction and grounding methods, Detection of faults and abnormal operating conditions. [4 Lectures]
- **Distribution system protection:** Feeder protection philosophies, Coordination examples, Power system restoration concepts - Reclosing, Automatic sectionalizing. [2 Lectures]
- **Introduction to relay setups and Standards:** Distance, overcurrent, over/under frequency relay set up demonstration. Discussion on standards for protection scheme in Indian power grid. [3 Lectures]

Text Books

1. S. Horowitz and A. G. Phadke, **Power System Relaying**, 4th Edition, Wiley, 2014.
2. A. G. Phadke and J. S. Thorpe, **Computer Relaying for Power Systems**, Wiley, 2009
3. Juan Gers, **Protection of Electricity Distribution Networks**, 3rd Edition, IET press, 2011.

References

1. P.M. Anderson, **Power System Protection**, Wiley-IEEE press, 1999.
2. J. L. Blackburn, T. J. Domin, **Protection Relaying: Principles and Applications**, 3rd Edition, Taylor and Francis, 2006
3. J.D. Glover, M. S. Saema, T. J. Overbye., **Power System Analysis and Design**, 5th Edition, Cengage Learning, 2010.
4. Bhavesh Bhalja, R. P. Maheshwari and N. Chothani, **Protection and Switchgear**, 2nd Edition, Oxford University Press, 2019.